
and recurrent neural networks Exercise 3 - Learning Transfer

Question 1

² The effect of different learning transfer methods on model performance and comparison of the role of fine tuning is the aim of this question. In the following, the desired dataset and the necessary experiments to be performed are mentioned. First, the dataset model performance was analyzed. Then, by designing and implementing the aforementioned experiments, the effect of various parameters on the Show.

Dataset: **10-CIFAR**

To answer the questions in this exercise, we will use the 10-CIFAR dataset . To load the 10-CIFAR dataset ,

And use transforms in PyTorch .

From the datasets.torchvision module

Selected models:

– 18ResNet (pre-trained on ImageNet)

– 121DenseNet (pre-trained on ImageNet)

Experiments

(A) Part One: The Model as a Feature Extractor

In this section, you should use pre-trained models as a fixed feature extractor. All layers

Just replace and train a new batch.

Freeze the base model and

Work steps:

Load 121DenseNet .

and – 18ResNet models

– Freeze all convolution layers so they don't get gradients.

– Replace the classification layers with the following architecture:

– Train only these classification layers.

Transfer Learning1

Fine tuning2

Required report:

Train/Val/Test Accuracy–

Train/Val Loss Chart–

– Explain how well the features extracted from ImageNet fit 10–CIFAR without fine-tuning .

(b) Part Two: Fine-tuning the Models

In this section, you should examine how the gradual opening of layers affects performance.

Configuration:

– All layers are frozen.

- Only the last free block

- The last 3 free blocks

– Entire free model (doing this is a privilege.)

Work steps:

– Use the same category as the first section.

– Unfreeze the necessary blocks and train the network.

– Compare the results with the feature extraction mode.

Questions

Do you find it? – What effect does learning rate have on learning transfer performance? How to achieve the optimal amount?

– What effect does the number of frozen layers have on model performance?

– How does model architecture affect learning transfer? Compare the base models in each case.

– How effective has transfer learning been in increasing accuracy?

Note that to answer each question, you must design and implement the necessary experiments and analyze the results.

Question 2

The purpose of this question is to examine the performance of recurrent models³ in predicting time series and analyze their sensitivity to the size of the latent state and the length of the sequence. In this exercise, after loading the dataset, you must design, train different models, and analyze their results.

Dataset: Electricity consumption

We will use the Electricity Load Diagrams dataset to include electricity consumption measurements in this question.

You can use the torchts package or any other standard method to load this dataset at consecutive time intervals.

And the full description of the dataset can be viewed at the following link: <https://archive.ics.uci.edu/dataset/321/electricityloaddiagrams20112014>.

Main task: Predict the amount of electricity consumption in the next time step using past values.

Selected models:

Elman RNN –

LSTM –

GRU –

(A) Part One: Comparison of Linear Model with Recursive Model

In this section, you should compare the performance of a simple linear model with a recursive model.

Work steps:

– Create a linear regression model whose input is the last sequence and whose output is the next value.

– Design and train an Elman RNN model with a tanh activation function .

– Evaluate both models on Test/Val/Train .

Questions:

trial and error. Evaluate the performance of the models. – Find and report the best hyperparameters for each model through

Use the MSE evaluation criterion .

– Plot the prediction graph of the linear model and the RNN model separately. Each graph should include the actual data in

Plot one curve and the model prediction on another curve. Note that this plot should only include the test data.

– Which model performed better? Why?

Recurrent Neural Networks³

GRU and (b) Part Two: LSTM Comparison

In this section, you should compare two common recursive architectures under identical conditions.

Fixed configuration:

- Hidden size = 32
- Sequence length = 20
- A recurrent layer

Work steps:

Design and train the GRU . and – LSTM models

– Compare the training process and the MSE Test value.

– Draw the prediction graph of both models.

Questions:

– Investigate which model is more stable and what is the possible reason for this behavior.

(c) Part Three: Sensitivity Analysis of Regression Models

experiments: In this section, you should examine the sensitivity of three key hyperparameters through systematic

- Hidden size
- Sequence length
- Recursive model type

Experimental values:

Hidden size: {64, 32, 16,}

Sequence length: {100, 50, 20,}

Model used: {Elman, LSTM, GRU{

Work steps:

- Run the full training once for each possible combination.
- Draw at least three diagrams of the training process (as examples).
- Prepare a comparison table of the performance of the models.

Necessary analysis:

- Based on the results of the experiments, analyze what effect increasing or decreasing each hyperparameter has on the performance of the model.

Explanations

1 - All implementations must be done with PyTorch .

2- The final report must include tables, graphs, and analyses.

3 - Standard data division: .70/20/10

.StudentID_HW03.zip : Final file - 4

5 - Every day of delay will result in a 10% reduction in grade.

6 - To ask a question to the TA:

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